Kredo: A Contribution-Based Mutual-Credit Economy with Soulbound Reputation and Compensatory Emission

Fedor Chishchin
Independent Researcher
fedorstartup@gmail.com

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Abstract

We present *Kredo*, an economic mechanism designed to reward the creation of value rather than the position of ownership. The design is derived top-down: five philosophical axioms are formalized as seven state invariants, and every transition function is required to preserve them. The economy uses two assets: a transferable contribution token V (working money and a claim on the club's external profit) and a non-transferable "soulbound" reputation token R. Credit is issued at the moment of a confirmed transaction and accompanied by a *compensatory emission*—a deferred, escrowed "dividend to all" whose magnitude is governed by a dynamic coefficient ε that shrinks as the observed default rate grows. We prove that this coefficient keeps the money supply bounded relative to verified value under a nonzero default rate (Theorem 1), and that operations against the liquidity fund are price-neutral (Theorem 2), so the token's price responds only to fundamentals. Governance uses quadratic ($\sqrt{\mathrm{R}}$) voting under a dual reputation-and-stake quorum, which, together with the soulbound property of R, makes capture by capital or by Sybil accounts economically unprofitable. We derive a minimal external-revenue *survival condition* for a non-declining price, show that the discounted-withdrawal rule leaves the fund positive at every level of exit, and prove a stake threshold of order $\bar{N}/a$ (mean transaction size over audit rate) beyond which even the optimal adaptive wash-trading fraud is loss-making. A deterministic agent-based simulator implementing the full mechanism is exercised across 315 stability runs—Monte Carlo, a 192-point parameter sweep, and combined-attack stress tests—none of which produced an economic collapse, while a separate regime analysis reproduces the predicted price trajectories. All code, derivations, and results are released openly.

Keywords: mutual credit; complementary currency; mechanism design; token economics; quadratic voting; Sybil resistance; agent-based simulation.

1 Introduction

Two widely deployed classes of economic organization each embed a structural unfairness. In *rent-based capitalism*, the holder of capital earns without contributing: value is extracted from a position of ownership rather than created, and the producer of value and the recipient of profit are frequently distinct agents. In *egalitarian cooperatives and communes*, all members are treated equally, but the absence of a growth-and-reward mechanism yields the classical free-rider problem: contribution is not differentially rewarded, active members bear disproportionate cost, and the system stagnates.

This paper studies a mechanism, *Kredo*, intended as a third option: reward proportional to the creation of value, in which active members earn more than passive ones yet no member is left with nothing, and in which the system defends itself against inflation, panic, and fraud without discretionary intervention.

Our contributions are as follows.

1. A top-down design method: five axioms (Section 3.1) are formalized as seven invariants that every transition function must preserve (Section 3.3), yielding a specification in which the intended ethics are machine-checkable properties.
2. A compensatory-emission mechanism with a dynamic coefficient ε, and a proof that it bounds supply relative to verified value under nonzero defaults (Theorem 1).
3. A pricing rule tied strictly to fundamentals, with a proof that fund operations are price-neutral (Theorem 2) and that the discounted-withdrawal rule cannot exhaust the fund under a run (Theorem 3).
4. A governance design combining $\sqrt{\mathrm{R}}$ voting with a dual quorum and a soulbound reputation token (Proposition 1), and a characterization of exactly when wash-trading fraud—including the optimal adaptive strategy—is unprofitable (Theorems 4 and 5).
5. A minimal external-revenue survival condition with the long-run price limit (Propositions 2 and 4), a consolidated set of deployment conditions (Section 11), and an empirical validation over 315 simulated runs.

All artifacts—the mechanism, its derivations, the simulator, the experiment protocols, and the raw results—are publicly available.[1]

2 Related work

The idea that money should not accrue rent to idle holders traces to Gesell's proposal of a carrying cost on currency [1]. Mutual-credit systems, in which the money supply is created endogenously by bilateral trade and nets to zero, have a long practical history [2]; empirical work on the Swiss WIR/*Wirtschaftsring* documents their counter-cyclical, macro-stabilizing behavior [3]. Contemporary blockchain-based complementary currencies such as Circles UBI [4] and GoodDollar [5] distribute newly issued units broadly; Kredo's "dividend to all" is a deferred, contribution-weighted, default-conditioned variant of this idea. Trustless issuance and the avoidance of pre-mine follow the design ethos of permissionless value systems [6]. Our governance draws on quadratic voting [7] and quadratic funding [8], whose square-root weighting limits plutocratic dominance; the soulbound (non-transferable) treatment of reputation is what makes the quadratic weighting robust to vote-buying. Resistance to Sybil attacks is analyzed in the tradition of Douceur [9]. Finally, we evaluate the mechanism with agent-based simulation [10], the standard tool for studying emergent behavior of many interacting agents when closed-form analysis is intractable.

3 The model

3.1 Axioms

Axiom 1 (Value is created, not extracted)**.** Reward must correspond to the creation of value, not to a position of ownership.

[1] https://github.com/fedorello/kredo-model

Axiom 2 (Value is subjective but consensual)**.** A fair price is the one two parties agree upon voluntarily under the availability of alternatives.

Axiom 3 (Symmetry of gain and loss)**.** All participants are structurally bound to the fate of the system; no class is shielded from the risks of the others.

Axiom 4 (No privileged knowledge)**.** All rules, balances, and decisions are transparent.

Axiom 5 (Power from contribution)**.** Influence over the rules follows from demonstrated useful participation, not from accumulated capital.

3.2 State

Definition 1 (System state)**.** At discrete time t the state is the tuple

$$S_t = (M_t,\ b_t,\ r_t,\ d_t,\ F_t,\ E_t,\ G_t,\ H_t),$$

where M_t is the finite set of members; $b_t : M_t \to \mathbb{R}$ the V-balance (which may be negative up to a credit limit); $r_t : M_t \to \mathbb{R}_{\geq 0}$ the reputation; $d_t : M_t \to \mathbb{R}_{\geq 0}$ accumulated debt; $F_t \in \mathbb{R}_{\geq 0}$ the liquidity fund (in an external stable asset, USDC); $E_t \in \mathbb{R}_{\geq 0}$ the distribution escrow; $G_t \in \mathbb{R}_{\geq 0}$ the genesis pool; and H_t the append-only transaction history.

3.3 Invariants

The seven invariants below formalize Axioms 1–5. Let $N_{\text{credit}}(\tau)$ be the newly issued (credit) portion of transaction τ, admitted only when its trust score confidence$(\tau) \in [0, 1]$ (Section 4) clears the gate of (I2); write $C(\tau) = N_{\text{credit}}(\tau)$ for this *confirmed contribution.* Write

$$\text{Supply}(t) = \sum_m \max\bigl(0, b_t(m)\bigr) + E_t + G_t, \qquad \text{VerifiedValue}(t) = \sum_{\tau \in H_t} \bigl(1+\varepsilon(\tau)\bigr) C(\tau) + \Gamma_t - \text{Burned}(t),$$

where $\varepsilon(\tau)$ is the compensation coefficient of (6) fixed at the time of τ, Γ_t is the cumulative genesis grant (itself funded by realized external revenue and fines, Section 5), and Burned(t) is the cumulative amount removed by defaults, penalties, and conversions. Since Supply contains issued principal, escrowed compensation, and genesis grants, all net of burns, VerifiedValue is defined to match it; any residual gap in (I3) reflects only the lag between fraud and its detection. (The reference implementation's stricter checker additionally weights $C(\tau)$ by confidence(τ); the resulting discount is a second, small source of the tolerated drift, since most transactions clear the auto-check at confidence = 1.)

(I1) Conservation on transfers. For any pure transfer, $\sum_m b_t(m) = \sum_m b_{t+1}(m)$.

(I2) Gated emission. V is minted only within the confirmation protocol and only if confidence$(\tau) \geq \theta_{\min}$.

(I3) Supply tracks value. $\bigl|\text{Supply}(t) - \text{VerifiedValue}(t)\bigr| \leq \epsilon_{\text{tol}}\, \text{Supply}(t)$, with $\epsilon_{\text{tol}} = 0.05$ absorbing detection lag.

(I4) Credit limit. $b_t(m) \geq -L\bigl(r_t(m)\bigr)$ for all m, t.

(I5) Fundamental price. $P_t = \bigl(F_t + \mu\, \text{ExtRev}_t\bigr) / \,\text{Supply}(t)$ is always computed, never assigned.

(I6) Soulbound reputation. No transfer map for R exists; R is only minted (for merit) and burned (for violations).

(I7) Universal rules. Every function depends only on observable state (b, r, d), not on member identity.

The admissible transition functions are Join, Leave, Transact, Repay, Default, Convert, Invest, together with derived periodic operations; each is required to preserve (I1)–(I7).

4 Credit and confidence

4.1 Credit limit

$$L(r) = L_0\big(1 + \alpha \ln(1+r)\big), \qquad L_0 = 100 \text{ V}, \quad \alpha = 0.5. \tag{1}$$

The logarithm bounds exposure: a member with $r = 10^4$ borrows at most ≈ 561 V, only $\approx$5.6× a newcomer's limit (Table 1). No single agent can therefore threaten solvency by defaulting.

r	0	1	10	100	1000	10000
$L(r)$	100	135	220	331	445	561

Table 1: Credit limit (1) as a function of reputation, rounded to the nearest integer.

4.2 Confidence

Axiom 2 forbids the mechanism from setting prices; instead it estimates how much a declared price can be trusted before minting against it. For a transaction $\tau = (A, B, N, c)$ with provider A, receiver B, declared value N, and category c,

$$\text{confidence}(\tau) = w_a s_a(\tau) + w_r s_r(\tau) + w_x s_x(\tau), \qquad w_a + w_r + w_x = 1. \tag{2}$$

The *auto-score* compares N to the category's recent price distribution. With sample mean μ_c and standard deviation σ_c over a 90-day window and $z = (N - \mu_c)/\sigma_c$,

$$s_a(\tau) = \begin{cases} 1, & |z| \le 1, \\ \exp\left(-\dfrac{(|z|-1)^2}{2}\right), & |z| > 1. \end{cases} \tag{3}$$

Thus $s_a = 0.61,\ 0.14,\ 0.01$ at $z = 2, 3, 4$. The *review-score* s_r is the median vote of randomly drawn reviewers (robust to outliers), and the *audit-score* s_x is 1 unless a stochastic audit finds a violation. Review is triggered when $|z| > 1$, when N exceeds the auto-approval limit $N_{\text{auto}} = 50\sqrt{r_A r_B + 1}$, when a party is under monitoring, or when the 3% random audit selects τ. Emission proceeds only when $\text{confidence}(\tau) \ge \theta_{\min} = 0.6$, per invariant (I2).

4.3 Market health and anti-concentration

Axiom 2 locates dishonesty not in a "wrong" price but in the absence of alternatives; accordingly the mechanism monitors structural market health rather than individual prices. For each service category it tracks a Herfindahl–Hirschman concentration index, the bilateral concentration of a member's counterparties, and short-cycle reciprocity (a signature of wash trading). When a category or cluster is flagged, the response is graduated and non-price: reduced credit limits for the implicated cluster, an elevated audit rate, suspension of compensatory emission on suspect transactions, and, in the limit, escalation to a community vote. The mechanism thus defends the *conditions* of honest exchange—transparency and the availability of alternatives—rather than dictating terms.

5 Emission and the compensation coefficient

5.1 Mechanics

In Transact, the credit portion is $N_{\text{credit}} = \max\big(0,\ N - \max(0, b_B)\big)$; the provider is credited N, and B's balance decreases by N, opening a debt of N_{credit} when $b_B < N$ (a pure transfer when $N_{\text{credit}} = 0$). Simultaneously a *compensatory emission* enters escrow,

$$\Delta E = \varepsilon(t)\, N_{\text{credit}}. \tag{4}$$

The escrow is released pro rata upon repayment and *burned* upon default. Released amounts are distributed across members in proportion to active turnover, where

$$\text{Turnover}(m,t) = \sum_{\tau:\ t-\tau<90d,\ m\in\tau} N(\tau)\ \text{confidence}(\tau),$$

via

$$\text{Share}(m,t) = \frac{\text{Turnover}(m,t) + \xi}{\sum_{m'} \text{Turnover}(m',t) + n_t\,\xi}, \qquad \xi = 1, \tag{5}$$

so the "dividend to all" rewards active contribution rather than idle holdings. The coefficient is dynamic in the observed default rate δ_t:

$$\varepsilon(t) = \max\Big(0,\ \min\big(\varepsilon_{\max},\ K^* - 1 - \kappa\,(\delta_t - \delta^*)\big)\Big), \tag{6}$$

with defaults $K^* = 1.5$, $\delta^* = 0.05$, $\kappa = 2$, $\varepsilon_{\max} = 0.95$. Hence ε ranges from 0.60 at $\delta = 0$ down to 0 at $\delta = 0.30$.

5.2 Stability under defaults

Theorem 1 (Bounded supply under defaults)**.** *Consider a stationary regime with constant default rate $\delta \le \frac{1}{2}$ and ε given by (6). Let C_{tot} be the total confirmed contribution $\sum_\tau C(\tau)$ and $P_{\text{net}} = (1-\delta)\,C_{\text{tot}}$ the net contribution surviving defaults. If defaulted issuance—both its principal and its compensation—is removed on remediation (the* Default *and fraud-remediation operations), then, up to the bounded genesis pool,*

$$\frac{\text{Supply}}{P_{\text{net}}} \le K^*.$$

Proof. Per unit of confirmed contribution the mechanism mints one unit of principal and, upon repayment, releases ε units of escrowed compensation, for $(1+\varepsilon)$ units of surviving supply; for a defaulted unit the compensation is burned and the principal removed on remediation, contributing nothing. With a fraction δ defaulting, $\text{Supply} = (1-\delta)C_{\text{tot}}(1+\varepsilon)$ while $P_{\text{net}} = (1-\delta)C_{\text{tot}}$, so $\text{Supply}\,/P_{\text{net}} = 1+\varepsilon$. By (6), $\varepsilon \le K^* - 1$, giving $\text{Supply}\,/P_{\text{net}} \le K^*$; for $\delta > \delta^*$ the term $-\kappa(\delta-\delta^*)$ makes ε strictly smaller, tightening the bound. □

Remark 1. Theorem 1 is the inflation brake: as realized defaults rise, ε falls automatically, and at $\delta = 0.30$ compensation is disabled entirely (strict conservation). The system does not assume full repayment; it bounds the money-to-value ratio for any $\delta \le \frac{1}{2}$. The remediation hypothesis isolates the adversarial (fraud) case; a good-faith default instead retains principal backed by the work actually delivered, which then counts toward net value, so the bound continues to hold (formalized without any remediation hypothesis in Proposition 3).

The genesis grant to a joining member, $g_0 = \min\big(100,\ G_t/\mathbb{E}[\text{new members}]\big)$, is likewise self-limiting: it contracts if growth outpaces the pool's inflow, so no subsidy is created *ex nihilo*.

5.3 Debt resolution and reactivation

Overdue debt triggers graduated sanctions rather than exclusion: at 30, 60, and 90 days the debtor faces escalating reputation decay and, ultimately, a freeze (balance and reputation reset, escrow burned) instead of expulsion. A frozen member may reactivate by repaying the outstanding debt and completing a probationary period restricted to small transactions. This regenerative treatment follows from Axiom 1: the objective is to restore productive participation, not to punish.

6 Reputation and governance

Reputation accrues through concave (activity) and linear (quality) terms and burns for failures:

$$\Delta \mathrm{R} = \beta_1 \ln(1+n_{\mathrm{tx}}) + \beta_2\, a_{\mathrm{audit}} + \beta_3 \ln(1+\mathrm{tenure}/30) + \beta_4\, n_{\mathrm{disputes}}, \tag{7}$$

$$\Delta \mathrm{R}^- = \gamma_1 f_{\mathrm{review}} + \gamma_2 f_{\mathrm{dispute}} + \gamma_3\, \mathrm{DefaultPenalty} + \gamma_4\, \mathrm{Fraud}, \tag{8}$$

with $\gamma_4 = 100$: proven fraud erases years of reputation. Concavity in bulk actions prevents reputation farming by volume.

Voting power is $\sqrt{\mathrm{R}(m)}$. Strategic decisions require a *dual quorum*: a reputation quorum $\sum_{\mathrm{voters}} \sqrt{r} \,/\, \sum_M \sqrt{r} > \theta$ and a stake quorum $\sum_{\mathrm{voters}} \max(0,b) \,/\, \sum_M \max(0,b) > \theta$, with $\theta = 0.5$ (0.66 for constitutional changes). Because R and V are distributed differently, the dual quorum blocks capture by activists and by whales simultaneously. A distinguished set of provisions is *constitutional*—the seven invariants, the soulbound property of R, and the prohibitions on pre-mine, founder allocations, and class-specific rules—and is amendable only by super-majority referendum, never by ordinary parameter governance.

Proposition 1 (Capital and Sybil resistance)**.** *Under invariant (I6), an attacker with arbitrary capital obtains zero reputation votes, since* R *is non-transferable. A Sybil attacker operating k accounts of base reputation r_0 obtains $k\sqrt{r_0}$ reputation votes; to reach a target of q votes requires $k = q^2/r_0$ accounts. At $r_0 = 0.5$, $q = 100$ votes require 2×10^4 accounts, each of which must clear identity screening and perform genuine activity to earn any* R*.*

Proof. The first claim is immediate from (I6): capital purchases V but no R, and voting weight is a function of R only. For the second, additivity of $\sqrt{\mathrm{R}}$ across identically endowed accounts gives $k\sqrt{r_0}$; setting $k\sqrt{r_0} = q$ yields $k = q^2/r_0$. Substituting $r_0 = 0.5$, $q = 100$ gives $k = 2 \times 10^4$. □

7 Pricing and the liquidity fund

The price is fixed by (I5), $P_t = (F_t + \mu\,\mathrm{ExtRev}_t)/\,\mathrm{Supply}(t)$ with $\mu = 12$ (a price-to-earnings multiple capitalizing annual external revenue).

Theorem 2 (Neutrality of fund operations)**.** Invest *(deposit ΔU of USDC for $\Delta V = \Delta U/P_t$ of* V*) and* Convert *(sell ΔV of* V *for $\Delta U = \Delta V\, P_t$) leave the price unchanged: $P_{t+1} = P_t$.*

Proof. For Invest, $F_{t+1} = F_t + \Delta U$ and $S_{t+1} = S_t + \Delta U/P_t$, where $S =$ Supply. Using $F_t = P_t S_t - \mu\,\mathrm{ExtRev}_t$,

$$P_{t+1} = \frac{F_t + \Delta U + \mu\,\mathrm{ExtRev}_t}{S_t + \Delta U/P_t} = \frac{P_t S_t + \Delta U}{S_t + \Delta U/P_t} = \frac{P_t(P_t S_t + \Delta U)}{P_t S_t + \Delta U} = P_t.$$

The argument for Convert is symmetric, with $F_{t+1} = F_t - \Delta V\, P_t$ and $S_{t+1} = S_t - \Delta V$. □

Thus price responds only to fundamentals (F and ExtRev); no fund operation manipulates it. A credit emission, by contrast, dilutes the price in the moment, $P_{t+1} = P_t\, S_t / \big(S_t + N_{\text{credit}}(1+\varepsilon)\big) < P_t$, and is recovered only by growth in ExtRev. Consequently internal turnover alone creates no external wealth—the mechanism's central honest conclusion.

Bank-run resilience. Let coverage $\rho_t = F_t/(P_t S_t)$. When $\rho_t < \rho^* = 0.3$, conversions are queued and discounted, $P_{\text{actual}} = P_t \min(1, \rho_t/\rho^*)$. The discount both protects the fund and incentivizes counter-cyclical investment at the bottom, restoring ρ; it also imposes a structural emission ceiling $\Delta S_{\max} = (F - F_{\min})/P_{\text{target}}$.

Profit distribution. External profit realized by the club is split each period: a majority is retained in the fund (raising P through (I5)), an activity-weighted dividend is paid to members, a share replenishes the genesis pool, and a small share funds operations. Moderators, arbiters, founders, and investors are all paid in V on identical terms—no class holds a claim senior to, or insulated from, the fate of the token. This is the operational content of Axiom 3: when the club gains, holders gain together; when it loses, they lose together.

8 Stationarity and the survival condition

Price is stationary when supply growth matches asset growth, $\dot{S}/S = \dot{\mathcal{A}}/\mathcal{A}$ with $\mathcal{A} = F + \mu\,\text{ExtRev}$. This partitions dynamics into a growth regime, a stable regime, and a falling regime.

Proposition 2 (Survival condition)**.** *A non-declining price requires an external-revenue growth rate satisfying*

$$\lambda_{\text{rev}} \;\geq\; \frac{1}{\mu}\Big(\text{EmissionRate} \cdot P - \lambda_{\text{inv}}\Big),$$

where λ_{inv} is the investment inflow rate.

Sketch. Since $P = \mathcal{A}/S$, differentiation gives $\dot{P} = (\dot{\mathcal{A}} - P\dot{S})/S$, so a non-declining price ($\dot{P} \geq 0$) is equivalent to $\dot{\mathcal{A}} \geq P\,\dot{S}$. Substituting $\dot{\mathcal{A}} = \lambda_{\text{inv}} + \mu\lambda_{\text{rev}}$ and $\dot{S} = \text{EmissionRate}$ (net of burns) yields $\lambda_{\text{inv}} + \mu\lambda_{\text{rev}} \geq \text{EmissionRate} \cdot P$; solving for λ_{rev} gives the stated bound. □

For instance, at $S = 10^5$, $P = 1$, $\mu = 12$, emission 5000 V/month and investment 2000 USDC/month, the club must earn at least 250 USDC/month externally. A permanently falling price in the absence of external revenue is therefore a mathematical necessity, not a design defect.

9 Further theoretical results

The results below strengthen Theorem 1 by removing its remediation hypothesis, characterize the long-run price, show that the discounted-withdrawal rule cannot exhaust the fund except in the limit of complete exit, prove that fraud is unprofitable at scale, and characterize the parameter regime that deters even the optimal adaptive fraud.

9.1 Stability without the remediation hypothesis

Proposition 3 (Generalized bound under mixed defaults)**.** *Partition issuance in a stationary regime into repaid credits (fraction $1-\delta$), good-faith defaults (fraction δ_g; the service was delivered, so the principal is retained and value-backed while the escrow is burned), and detected fraud (fraction*

δ_f*; principal and compensation burned), with* $\delta = \delta_g + \delta_f$*. Counting delivered work as value, set* $P_{\text{net}} = (1-\delta_f)\,C_{\text{tot}}$*. Then, with no assumption on the remediation of good-faith principal,*

$$\frac{\text{Supply}}{P_{\text{net}}} \;\le\; 1+\varepsilon \;\le\; K^*.$$

Proof. A repaid unit contributes $1+\varepsilon$ to supply, a good-faith default 1 (principal retained, escrow burned), and a detected fraud 0; hence Supply $= C_{\text{tot}}\big[(1-\delta)(1+\varepsilon)+\delta_g\big]$. Using $\delta - \delta_f = \delta_g$,

$$\text{Supply} - (1+\varepsilon)P_{\text{net}} = C_{\text{tot}}\big[(1-\delta)(1+\varepsilon)+\delta_g-(1+\varepsilon)(1-\delta_f)\big] = C_{\text{tot}}\big[\delta_g-(1+\varepsilon)\delta_g\big] = -\,C_{\text{tot}}\,\varepsilon\,\delta_g \;\le\; 0.$$

Thus Supply $\le (1+\varepsilon)P_{\text{net}}$, and $\varepsilon \le K^*-1$ by (6) gives the bound. Setting $\delta_g = 0$ recovers Theorem 1. □

9.2 The long-run price

Proposition 4 (Long-run price)**.** *Suppose the asset aggregate and supply grow at constant rates* $g = \lambda_{\text{inv}} + \mu\lambda_{\text{rev}}$ *and* $e = \text{EmissionRate} > 0$ *(net of burns), so* $\mathcal{A}(t) = \mathcal{A}_0 + gt$ *and* $S(t) = S_0 + et$*. Then* $P(t) = \mathcal{A}(t)/S(t)$ *is monotone and*

$$P(t) \xrightarrow[t\to\infty]{} \frac{g}{e} = \frac{\lambda_{\text{inv}} + \mu\lambda_{\text{rev}}}{\text{EmissionRate}}.$$

Moreover P *is non-decreasing iff* $P_0 \le g/e$*, which is exactly the survival condition of Proposition 2.*

Proof. $P'(t) = \dfrac{g\,S(t) - e\,\mathcal{A}(t)}{S(t)^2} = \dfrac{gS_0 - e\mathcal{A}_0}{S(t)^2}$, whose sign equals that of $g/e - \mathcal{A}_0/S_0 = g/e - P_0$ and is constant in t; hence P is monotone. As $t\to\infty$, $(\mathcal{A}_0+gt)/(S_0+et) \to g/e$. Finally $P_0 \le g/e$ rearranges to $\lambda_{\text{inv}} + \mu\lambda_{\text{rev}} \ge \text{EmissionRate}\cdot P_0$. □

9.3 The fund is not exhausted by a run

Theorem 3 (No fund exhaustion)**.** *During a withdrawal run conducted entirely in the discounted regime (*$\rho < \rho^*$*,* ExtRev *constant, no emission), in the continuous limit the fund and supply obey*

$$\frac{F}{F_0} = \Big(\frac{S}{S_0}\Big)^{1/\rho^*}.$$

Consequently $F > 0$ *for every* $S > 0$*: the fund is depleted only in the limit of complete exit (*$S \to 0$*).*

Proof. Let x index the amount of V converted, so $dS/dx = -1$. In the discounted regime the payout is $P_{\text{actual}} = P\,\rho/\rho^*$; using $P = \mathcal{A}/S$ and $\rho = F/\mathcal{A}$ gives $P_{\text{actual}} = F/(\rho^* S)$, so $dF/dx = -F/(\rho^* S)$. Hence $dF/dS = F/(\rho^* S)$, a separable ODE with solution $\ln(F/F_0) = \rho^{*-1}\ln(S/S_0)$, i.e. $F = F_0(S/S_0)^{1/\rho^*}$. Since $1/\rho^* > 0$, F and S vanish together. □

Remark 2. Because the payout $P_{\text{actual}} = F/(\rho^* S)$ falls with F, a holder's marginal proceeds vanish as the run deepens, removing the incentive to keep converting; with the theorem, this is the mechanism's structural answer to a panic.

9.4 Unprofitability of fraud at scale

A wash-trading fraud cannot extract value without executing fake transactions, each exposed to the audit, which makes the detection probability *endogenous* to the size of the theft.

Theorem 4 (Fraud is unprofitable at scale)**.** *Suppose extracting value $W > 0$ requires at least $n = \lceil W/\bar{N} \rceil$ wash trades ($\bar{N}$ the mean transaction size), each audited independently with probability $a > 0$; that new balances cannot be converted during a lock-up of τ_{lock} periods, over which detection has per-period probability at least p; and that a detected cluster forfeits stake $L > 0$ (burned balances, genesis grant, and reputation). Writing*

$$q(W) = (1-a)^{W/\bar{N}}\,(1-p)^{\tau_{\text{lock}}}$$

for the probability of reaching conversion undetected, the expected profit satisfies

$$\mathbb{E}[\pi] \;\le\; q(W)\,W - \big(1 - q(W)\big)L.$$

Consequently (i) $\mathbb{E}[\pi] < 0$ *whenever* $q(W) < L/(W+L)$*; and* (ii) *since* $q(W)\,W \to 0$ *as* $W \to \infty$*,* $\mathbb{E}[\pi] \to -L$*, so every sufficiently large fraud is strictly loss-making.*

Proof. Extracting W needs $n \ge W/\bar{N}$ wash trades; escaping every audit has probability $(1-a)^n \le (1-a)^{W/\bar{N}}$, and escaping detection through the lock-up has probability at most $(1-p)^{\tau_{\text{lock}}}$, so converting undetected has probability at most $q(W)$. Bounding the gain by W on that event and the loss by L on its complement gives the inequality. Part (i) is a rearrangement. For (ii), $(1-a)^{W/\bar{N}}W \to 0$ since an exponential in W dominates a linear one, whence $q(W)W \to 0$ and $\mathbb{E}[\pi] \to -L$. □

Remark 3. Detection is therefore endogenous: a larger theft demands more fake transactions, each an independent chance of exposure, so the survival probability decays exponentially in the amount stolen while the prize grows only linearly. The lock-up forces the attacker to hold the unbacked position while this hazard accrues. The optimum over adaptive strategies (e.g. splitting the theft across many clusters) is characterized next, in Theorem 5.

9.5 Deterring the optimal adaptive fraud

Theorem 4 bounds a single cluster of fixed size; an adaptive adversary may instead split the attack across many clusters and choose each size. Because the audit and the forfeitable stake act *per cluster*, the adaptive problem separates. Write $\beta = \ln\!\big(1/(1-a)\big)/\bar{N} > 0$ and $c = (1-p)^{\tau_{\text{lock}}} \in (0,1]$, so a cluster stealing W against stake L has expected profit at most

$$g(W) = c\,e^{-\beta W}(W+L) - L,$$

with equality when the bounds of Theorem 4 bind (the defender's worst case).

Lemma 1 (Separability)**.** *For any adaptive strategy partitioning the attack into m clusters with independently audited transactions and independently forfeitable stakes, the total expected profit is at most $\sum_{i=1}^{m} g(W_i) \le m \max_{W\ge 0} g(W)$. Concentration monitoring, which raises detection for coordinated clusters, only decreases the left-hand side.*

Proof. Independence makes the per-cluster bounds of Theorem 4 additive, and each summand is at most $\max_W g$. Monitoring introduces positive detection dependence across a coordinated cluster set, lowering each q_i and hence each $g(W_i)$. □

Theorem 5 (Deterrence of the optimal adaptive fraud)**.** *Every adaptive strategy in the class of Lemma 1 has non-positive expected profit iff* $\max_{W\geq 0} g(W) \leq 0$. *With* $W^\dagger = 1/\beta - L$*, this holds iff either* $W^\dagger \leq 0$*, or* $W^\dagger > 0$ *and*

$$\frac{c}{e\,\beta}\,e^{\beta L} \;\leq\; L.$$

In particular, the stake condition

$$L \;\geq\; \frac{1}{\beta} = \frac{\bar{N}}{\ln\!\big(1/(1-a)\big)} \;\approx\; \frac{\bar{N}}{a}$$

is sufficient: it forces $W^\dagger \leq 0$*, whence* g *is decreasing with* $g(0) = (c-1)L \leq 0$.

Proof. By Lemma 1 the class optimum is $m \max_W g$, attained in the worst case by m clusters at the maximizer; hence every strategy is non-positive for every m iff $\max_W g \leq 0$. Now $g'(W) = c\,e^{-\beta W}\big(1 - \beta(W+L)\big)$ vanishes only at $W^\dagger = 1/\beta - L$; hence g increases then decreases when $W^\dagger > 0$ and is decreasing on $[0,\infty)$ when $W^\dagger \leq 0$. In the latter case $\max g = g(0) = (c-1)L \leq 0$. In the former, using $\beta W^\dagger = 1 - \beta L$ and $W^\dagger + L = 1/\beta$,

$$\max_W g = g(W^\dagger) = c\,e^{-\beta W^\dagger}(W^\dagger + L) - L = \frac{c}{e\,\beta}\,e^{\beta L} - L,$$

so $\max g \leq 0$ is the displayed inequality; $L \geq 1/\beta$ gives $W^\dagger \leq 0$. □

Remark 4. The condition is a concrete—and not free—design requirement. With the illustrative $a = 0.03$, $\bar{N} = 50$ V the threshold is $L \geq \bar{N}/\ln(1/(1-a)) \approx 1642$ V, whereas the default stake $L = 300$ V leaves a profitable optimum near $W^\dagger \approx 1342$ V with $g(W^\dagger) \approx +425$ V (compare Table 4, whose sampled maximum is 407 at $W = 1000$). Deterrence therefore requires the per-cluster forfeitable stake—reputation plus locked collateral—to reach $\Theta(\bar{N}/a)$, attainable by raising the audit rate a, lengthening the lock-up, or requiring collateral before large exposure. This settles the optimum for independently-detected clusters; a fully coordinated adversary that deliberately correlates clusters to defeat monitoring, and dynamic timing, remain for future work (Section 13).

10 Numerical illustration

We instantiate the closed-form results with the paper's default parameters; the deterministic simulator of Section 12 reproduces the same qualitative behavior across its 315 runs.

Long-run price (Proposition 4). With $\mu = 12$, net emission $e = 5000$ V/month, investment $\lambda_{\text{inv}} = 2000$ USDC/month and initial price $P_0 = 1$, the limit $g/e = (\lambda_{\text{inv}} + \mu\lambda_{\text{rev}})/e$ maps external revenue directly onto the three regimes (Table 2): the stable case reproduces the survival threshold $\lambda_{\text{rev}} = 250$ exactly.

λ_{rev} (USDC/mo)	0	250	500
limit g/e (USDC/V)	0.40	1.00	1.60
regime	C (falling)	B (stable)	A (growth)

Table 2: Long-run price versus external revenue.

Fund non-exhaustion (Theorem 3). With $\rho^* = 0.3$ the closed form $F/F_0 = (S/S_0)^{1/\rho^*}$ leaves the fund strictly positive at every level of exit (Table 3); even a 90% simultaneous exit leaves it nonzero.

Exit	10%	30%	50%	70%	90%
S/S_0	0.90	0.70	0.50	0.30	0.10
F/F_0	70.4%	30.5%	9.9%	1.8%	0.05%

Table 3: Fund coverage retained after a simultaneous exit.

Unprofitability at scale (Theorem 4). With per-transaction audit $a = 0.03$, mean size $\bar{N} = 50$ V and stake $L = 300$ V, the audit factor $(1-a)^{W/\bar{N}}$ alone pushes the expected-profit bound below zero once the theft is large (Table 4); the lock-up factor $(1-p)^{\tau_{\text{lock}}}$ only lowers it further.

W (V)	500	1000	2500	5000	10000
wash trades n	10	20	50	100	200
survive audit q	73.7%	54.4%	21.8%	4.8%	0.2%
$\mathbb{E}[\pi]$ bound (V)	290	407	311	-48	-277

Table 4: Fraud expected-profit bound (audit factor only); it turns negative and tends to $-L$ as the theft grows.

11 Design guidance

The analysis yields a small set of parameter conditions a deployment must satisfy. We collect them here; each is derived above and is a lever the community can tune by governance.

Bounded inflation. Keep $\varepsilon \leq K^* - 1$ via (6), so Supply $/P_{\text{net}} \leq K^*$ (Theorem 1, Proposition 3). A smaller K^* tightens the money-to-value ratio. Default $K^* = 1.5$.

Non-declining price. Require external revenue $\lambda_{\text{rev}} \geq (\text{EmissionRate} \cdot P - \lambda_{\text{inv}})/\mu$; the price then tends to $(\lambda_{\text{inv}} + \mu\lambda_{\text{rev}})/\text{EmissionRate}$ (Propositions 2 and 4). The multiplier $\mu = 12$ and the investment inflow are the levers.

Fund solvency under a run. The discount engages at coverage $\rho < \rho^*$ and keeps $F = F_0(S/S_0)^{1/\rho^*} > 0$ for all $S > 0$ (Theorem 3); the emission ceiling $\Delta S_{\max} = (F - F_{\min})/P_{\text{target}}$ preserves buy-back capacity. Default $\rho^* = 0.3$.

Bounded single-agent exposure. The credit limit $L(r) = 100\,(1 + 0.5\ln(1+r))$ caps any one member's default loss (I4): at most 560 V even at $\mathrm{R} = 10^4$.

Gated emission. Mint only when confidence$(\tau) \geq \theta_{\min}$ (I2). Default $\theta_{\min} = 0.6$.

Capital and Sybil resistance. Vote by $\sqrt{\mathrm{R}}$ with R soulbound under a dual quorum θ; capital buys no votes, and q votes cost q^2/r_0 genuine accounts (Proposition 1). Defaults $\theta = 0.5$ (0.66 for constitutional changes).

Fraud deterrence. Ensure the per-cluster forfeitable stake $L \geq \bar{N}/\ln(1/(1-a)) \approx \bar{N}/a$ (Theorem 5). The levers are the audit rate a, the lock-up τ_{lock}, and required collateral. *The illustrative defaults ($a = 0.03$, $\bar{N} = 50$, $L = 300$ V) fall short of the threshold (≈ 1642 V) and must be strengthened.*

Of the illustrative defaults only the fraud-stake condition is unmet; the structural conditions hold by construction, and the price condition is a requirement on realized revenue rather than a parameter choice. Choosing K^*, μ, ρ^*, $\theta_{\min}$, and the stake/audit parameters within these bounds is the whole of a deployment's economic tuning.

12 Empirical validation

12.1 Method

We implemented the full mechanism as a deterministic agent-based simulator with an injected, seeded pseudo-random source, so that a seed reproduces a run bit-for-bit. Stability is graded by five criteria (V2): (C1) price does not reach zero; (C2) fund coverage $\rho \geq 0.05$ on at least 95% of ticks; (C3) at most 20% of members frozen; (C4) non-negative supply; (C5) invariants clean. We report (C1)–(C4) as *economically stable* separately from (C5), because the (I3) bookkeeping check—the implementation's stricter, confidence-weighted variant of (I3)—has a known drift during the auto-repay/escrow-distribution lifecycle. The protocol comprises: reproduction of reference scenarios (V1); Monte Carlo over 4 scenarios $\times$ 30 seeds (V3); a 192-point sweep over $(K^*, \mu, \delta^*, \rho^*)$ (V4); combined-attack stress tests (V5); and regime analysis (V6).

12.2 Results

Across 315 runs, no run produced an economic collapse (Table 5). The parameter sweep was stable at all 192 points, indicating the mechanism is not brittle to parameter choice. Combined fraud-plus-bank-run and 70% simultaneous-exit stress tests remained economically stable via the withdrawal queue and escrow burn, and the three predicted price regimes were reproduced.

Protocol	Runs	Economic collapses
Monte Carlo (V3)	120	0
Parameter sweep (V4)	192	0
Stress tests (V5)	3	0
Total	315	0

Table 5: Validation summary. Full raw outputs accompany the released code.

13 Limitations

Empirical stability is necessary but not sufficient. Our runs cover 1–3 months of virtual time; slow effects over multi-year horizons are untested. The agents are statistical rather than strategic, so adaptation and political organization by real participants are out of scope; the optimal fraud over independently-detected clusters is characterized in Theorem 5, but a fully coordinated adversary (correlating clusters to defeat monitoring) and dynamic timing are not tested empirically. Legal and fiscal questions—jurisdiction, identity/KYC, and the taxation of micro-transactions—lie outside the model. The mechanism therefore constitutes a structurally sound foundation whose behavior with real human participants remains to be established.

14 Conclusion

Kredo demonstrates that a coherent ethical stance—reward for creation, symmetry of fate, transparency, and power from contribution—can be compiled into a precise mechanism whose intended properties are provable and machine-checkable. The compensatory-emission coefficient bounds inflation under defaults, fund operations are price-neutral and the fund survives any partial run, governance resists both capital and Sybil capture, fraud at scale is loss-making once the forfeitable stake reaches the derived threshold, and a minimal external-revenue condition governs survival. A deterministic simulator corroborates structural resilience across 315 runs. The open release invites replication, adversarial analysis, and extension toward a small live deployment.

Data and code availability

The mechanism specification, full derivations, the simulator, the experiment protocols, and the raw results are openly available at `https://github.com/fedorello/kredo-model` under the MIT license.

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